



Decoding Fake News and Hate Speech: A Survey of Explainable AI Techniques

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This survey emphasizes the significance of Explainable AI (XAI) techniques in detecting hateful speech and misinformation/Fake news. It explores recent trends in detecting these phenomena, highlighting current research that reveals a synergistic relationship between them. Additionally, it presents recent trends in the use of XAI methods to mitigate the occurrences of hateful and Fake contents in conversations. The survey reviews state-of-the-art XAI approaches, algorithms, modeling datasets, as well as the evaluation metrics leveraged for assessing model interpretability, and thus provides a comprehensive summary table of the literature surveyed and relevant datasets. It concludes with an overview of key observations, offering insights into the prominent model explainability methods used in hate speech and misinformation detection. The research strengths, limitations are also presented, as well as perspectives and suggestions for future directions in this research domain.

CCS Concepts: • **Computing methodologies** → **Information extraction**; *Machine learning approaches*; • **Human-centered computing**; • **General and reference** → Surveys and overviews;

Additional Key Words and Phrases: Fake news, hate speech, explainable AI, interpretable AI, machine learning techniques

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1 Introduction

The rapid spread and democratization of **Artificial Intelligence (AI)** are driving most communities, and sectors, public or private, to adopt more AI-centric ideologies and modes of operation, following key takeaways from the recent [AI index report](https://aiindex.stanford.edu/ai-index-report-2023/).¹ At the dawn of the Fourth industrial revolution, we are witnessing a mass digital transformation and exponential growth in **Machine Learning (ML)**-based applications, whose main objectives are to accelerate business growth through automation, thus boosting market dynamics and overall customer satisfaction, and retention. The promises of AI are soaring, especially for streaming and social media companies that rely heavily on ML and very recently, sophisticated **Generative AI (GenAI)** Models to power tasks ranging from personalized social feeds, targeted advertising, and product recommendations to biometrics and cyber protection [1–2] through sophisticated data analysis methods to help identify and thus filter out inaccurate and hateful contents. Recent projections [3] estimate the global Market for “AI for social media” to skyrocket at a 28.3% Compound Annual Growth Rate, equivalent to an estimated 1.6-billion-dollar market bump from 2018. This exponential rise in digital platforms has facilitated human interactions by providing an alternative medium for people to connect, communicate, and exchange information. Hence, fostering basic human rights of freedom of expression and access to information, which on the flip side has also led to the increasing spread of misinformation [4]. This has, in turn, been known to materialize into grave social incivilities ranging from implicit slights [9] to more explicit slurs such as harmful/toxic rhetoric that can manifest into xenophobia, cyber harassment, and bullying as documented by [5–8]. The consequential negative and damaging effects of these abuses on our society, particularly those related to hate, have triggered an outburst of studies geared toward designing and deploying tools to automatically detect these forms of social incivilities in text or spoken conversations through the use and implementation of sophisticated AI, ML, and **Natural Language Processing (NLP)** Techniques. However, even with such unprecedented development and deployment of **State-Of-The-Art (SOTA)** technologies in detecting and mitigating these issues, a crucial limitation to their implementation for most real-world critical use cases is the lack of human understandability, and interpretability of their inner workings [10].

2 Research Background

2.1 Overview of Main Concepts

This research focuses primarily on two interconnected concepts: Fake news and Hate speech. Hate speech as described by Meta in 2024 [11] as a direct attack on individuals based on protected characteristics such as race, ethnicity, national origin, disability, religious affiliation, caste, sexual orientation, sex, gender identity, and severe disease. On the other hand, Fake news or Misinformation (as will be used interchangeably in this article) has been described as the sharing of known false information to the general public [12] or as fabricated information presented as the truth [13], drafted in a way to cause an emotional load, with aims to deceive, influence opinions to induce implicit conclusion [33], and may result in harm to a person’s or entity’s reputation. Despite its existence long before the data revolution [14], the 2016 US presidential election catalyzed the recent uptick of misinformation spread [15], amplified by digital and social media access and widespread use [16–17]. The captivating nature of these dubious news pieces generally made available to users through sophisticated content selection algorithms, recommendation systems, and their continuous re-shares and likes, have exacerbated this problem. Indeed, misinformation and fake news are major global concerns, particularly during elections

¹<https://aiindex.stanford.edu/ai-index-report-2023/>.

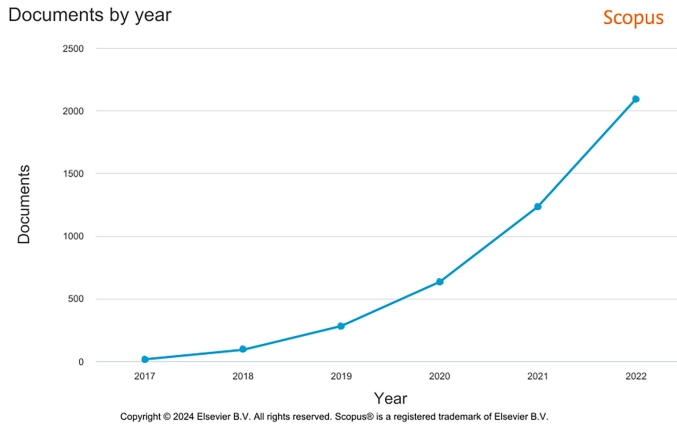


Fig. 1. Scopus research interest in XAI term from 2017 to Jan 2023.

and politically charged periods, as highlighted by a recent [UNESCO](#) report which underscores the widespread societal implications and fear surrounding these issues. The reports reveal that 85% of survey respondents expressed profound concerns about the ramifications of hate speech, while 66% were worried about the impact of fake news in their communities. This troubling statistic highlights the pervasive anxiety around these issues, which threatens to undermine democratic processes and social cohesion. Therefore, there is a pressing need for robust strategies to improve their detection and management, thus ensuring the preservation of public discourse integrity.

Interpretability by Miller (2019) [18] account is the degree to which humans understand the impact of a particular decision, while explainability is often associated with an AI system’s internal logic and inner workings. While Saeed and Omlin (2023) [19] have provided clear distinction to the use of both terms, the terms “interpretable” and “explainable” are often used interchangeably in the ML community, with explainability rapidly gaining mainstream acceptance [21]. Nonetheless, **Explainable AI (XAI)** was coined by Van Lent et al. (2004) [20] to describe the systems’ ability to explain the behavior of AI-driven units in simulation games. Thus, building on this foundational idea, the **Defense Advanced Research Projects Agency (DARPA)** [21] further elaborated on this definition, describing XAI, as any AI system capable of explaining their thought process to a human user such that they can understand the system’s strengths and weakness, and thus enable them to develop a knowledge of their future behaviors. The lack of transparency and the inability to understand the decision-making processes of most AI systems are the primary reasons for their limited use, adoption, and trust by humans, contributing to their reputation as “black boxes.” XAI aims to make AI system outputs more transparent, understandable, and interpretable by humans, and the need for XAI is vast, as extensively elaborated by [10], who highlights four key takeaways for their necessity. These include the fact that: -

- Explainability helps to interpret features.
- Explainability is necessary to mitigate Biases in AI systems as they can be instrumental in uprooting biased features in sentences.
- Explainability served as a verification and validation mechanism before deploying products to users and even after deploying, to ensure it works as intended.

Over the past five years, interest in XAI has grown significantly (See Figure 1), with a booming momentum in online searches and its application to diverse interdisciplinary fields [22–24]

This spike in XAI research has equally been met with a staggering 92% increase in research publications related to methods for automatically detecting social issues such as fake news, as discussed in [25], thus leading to a plethora of investigative research on ways to build reliable, responsible, and explainable tools for detecting such contents in everyday conversations. There has been a great deal of survey research related to XAI over the past years, with previous literature reviews focusing mainly on educating researchers about underlying concepts and taxonomies [26, 27], highlighting different XAI approaches and techniques often used across various implementation and use cases [23]. Some have pondered on the recent status, emerging trends, and future directions [28], generally through systematic [23, 29] and analytics perspectives [24]. Others, on the other hand, have emphasized the opportunities and challenges of XAI applicability in the real world [30, 31], and demonstrated the pros and cons [32] of current uses and implementation of XAI, with a consensus insisting on its relevance for ethical and responsible AI practice.

A recent SOTA review by [Gongane et al. \(2024\)](#) [10] highlights this need by focusing on XAI methods applied to fake news and hate speech, particularly on social media data. The review provides a comprehensive overview of various XAI methods, emphasizing the evaluation metrics used to assess the models' explainability in detecting hate speech and fake news. The work also provides technical and ethical challenges associated with using such XAI methods to comprehend the nuances of text shared on social media platforms.

Our work extends beyond [10] by exploring the broader application of these methods beyond social media data to also include references to News Media, Medical COVID-19 data, and Political content. We equally summarize the research findings, elaborating on the strengths and weaknesses of the proposed approaches, and offering valuable insights into the latest ML algorithms, evaluation metrics, and methodological considerations. An emphasis on the XAI methods and the Datasets used is made and highlights opportunities for future research and innovation. This review underscores the potential for XAI in detecting misinformation and hate speech, a perspective that aligns with our goal to explore them comprehensively, thereby contributing to the development of solutions to combat them more effectively.

2.2 Research Motivation

While fake news and hate speech may initially seem like distinct phenomena, they are often more intertwined than traditionally perceived. Typically studied in isolation, a recent article by [Gongane et al. \(2024\)](#) [10] demonstrates that Fake news and hate speech share a critical commonality: their intent to deceive and inflict harm. Moreover, a study by [Mosleh et al. \(2024\)](#) [8] demonstrates the necessity of considering these concepts together and proves their interconnectedness, especially in the context of online communication and social media. The authors' analysis shows a direct relationship between misinformation and harmful language by proving that a substantial percentage of fake news headlines incorporate hateful language. The study also reveals that users who engage with and retweet low-quality content are more likely to spread hate. Conversely, [Kim et al. \(2024\)](#) [25] offer a different perspective by investigating the exposure to misinformation among users who produce hate speech. Their findings indicate that individuals who frequently engage in hate speech are also more likely to encounter and be influenced by fake news. This dual approach underscores the intertwined nature of both concepts, suggesting that addressing one issue could inherently impact the other.

Our research ideology (See Figure 2) aligns with the perspective that fake news and hate speech are deeply interconnected and often amplify and influence each other. This belief is grounded in the following observation.

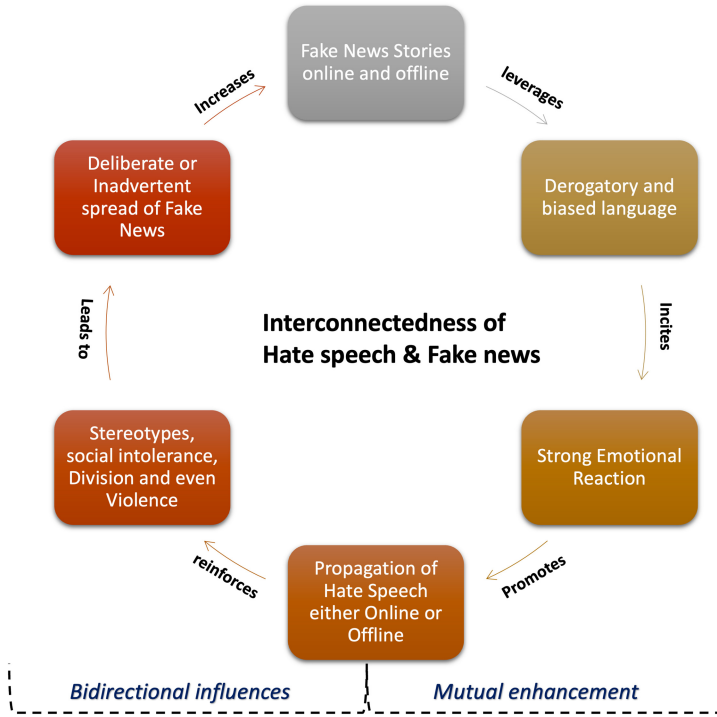


Fig. 2. Feedback loop illustrating the interconnected dynamics between fake news and hate speech.

- **Mutual enhancement:** Fake news propagators are more likely to spread hate [8], as their stories frequently use derogatory and biased language to provoke strong emotional reactions [33]. This can thus contribute to reinforcing stereotypes and fostering division online while potentially leading to violent actions offline [34], inciting political uprisings, and even amplifying domestic terrorism, as demonstrated in several studies [35–37].
- **Bidirectional influences:** Individuals who engage in hate speech may deliberately or inadvertently spread fake news. As shown in Kim et al. (2024) [25], those who produce hate speech are more likely to be exposed to and influenced by fake news. This makes them more susceptible to spreading falsehoods, either to further their agendas or due to their heightened exposure to misinformation or their overall ideological alignment. Such actions often target specific social demographics, aiming to incite hatred and violence. And the cycle continues.

Our work follows from Gongane et al. (2024) [10] and as such, we also believe in a synergistic relation, as demonstrated by [8, 10, 25]. We also believe that the interconnectedness of both concepts necessitates studying them in tandem as this will offer better and more profound insights into their dynamics and impact on society.

2.3 Research Methodology

The research methodology comprises articles extracted from popular databases including Web of Science (WoS), Google Scholar, and Scopus with the logic search including the title, abstract, and keywords for Scopus and WoS.

The logic search phrases used was: (((“explainable”) OR (“interpretable”)) AND (“fake news detection”) OR (“hate speech detection”)).

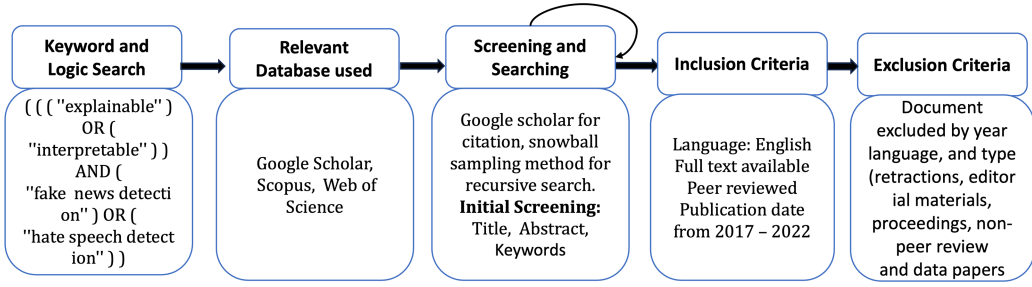


Fig. 3. Overview of research selection and inclusion criteria.

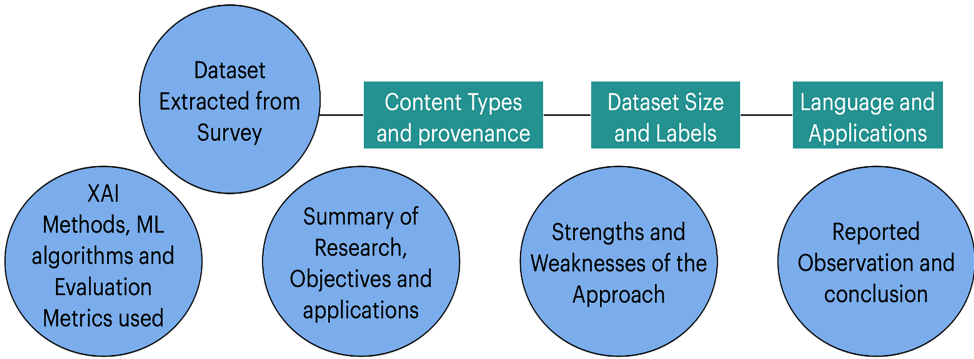


Fig. 4. Summary of Research Methodology and Deliverables4.

Additionally, we used a snowballing process to identify other relevant pieces of literature, for referencing and appropriate paper citing. Google Scholar was used for citation purposes and the retrieval of supplemental content. At the end of the research search and collection phase, the resulting manuscripts were grouped into clusters based on the category of offense and the similarity of approach to facilitate analysis and documentation of the research. Figure 3 provides a high-level representation of our research methodology and expected deliverables while Figure 4 shows a flow diagram of our data collection and categorization process. In Figure 5, we provide an overview of our methodology including our study's inclusion and exclusion criteria.

For each of the papers surveyed, we present the research objectives, applications, XAI methods, ML algorithms, and evaluation metrics used in the experiments. We also highlight the training datasets utilized, as well as the strengths and weaknesses of each approach, and report the research's observations and conclusions. A simplified tabular summary of the surveyed papers as well as a summary table of the surveyed datasets is presented in the supplemental material provided along with a summary table of the datasets extracted from the survey.

2.3.1 Research Questions. This research is guided by a set of Research Questions that align with our contributions:

- What XAI methods have recently been proposed and used for detecting fake news and hate speech in text? (Refer to Sections 3 and 4, respectively).
- How is the effectiveness of the proposed methods assessed in each surveyed paper? (Refer to Tables 1 and 2 from the supplemental material provided)

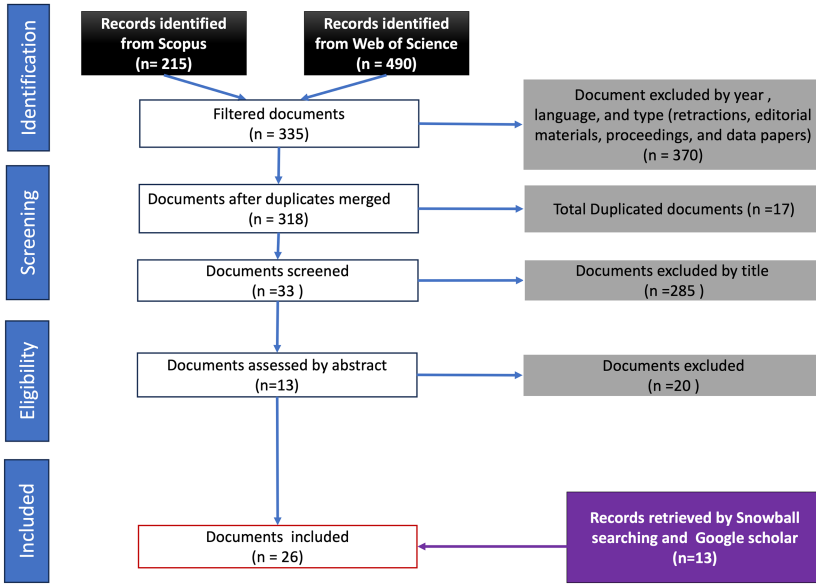


Fig. 5. Flow diagram of our research collection and categorization process.

- What are the most prominent trends in XAI research applied to hate speech and fake news? (See Section 5.1)
- What are the most commonly used datasets used in the study of hate speech and misinformation detection tasks? (See Section 5.2 and Table 1 in the supplemental material provided)
- What are the key findings, challenges, and successes as well as suggested future directions and research opportunities? (Refer to Sections 5.3 and 5.4).

We believe that answers to these research questions will be instrumental in gauging researcher interests in these concepts, facilitating and contributing to the growing bodies of studies in this area. We are confident this work will be essential in fast-tracking the conception and deployment of efficient and transparent applications for spotting and, hence, mitigating these issues in our communities.

2.3.2 Research Contribution. This Research contribution is summarized as follows:

- A comprehensive overview of XAI methods for detecting hateful and deceitful content is presented by
- Highlighting the ML algorithms implemented, the datasets, and the XAI method utilized in the research.
- A summary of the research core methodology, emphasizing the research evaluation metrics, and findings, as well as the strengths, and/or weaknesses of the proposed approach, is also emphasized.
- A highlight of key findings, the strengths, and limitations of our work, an outline of our future work, and plausible directions for further explorations on this topic, are presented.
- Finally, a tabular summary of the relevant datasets used from these tasks, highlighting their type, content, provenance, and specifications, such as label categories, applications, and the size is proposed in the supplemental material.

2.3.3 Research Organization. The remainder of this survey is organized as follows: Sections 3 and 4 explore the latest advances in XAI applications for detecting fake news and hate speech, respectively. In Section 5, we present key observations from the survey, discussing the strengths and limitations of our work. We also highlight the datasets identified in our survey that have been used for experimentation. As part of this discussion, we provide suggestions for future research opportunities related to these concepts. Section 6 concludes the survey. The supplemental material provided includes tabular summaries of the datasets used in the surveyed studies, detailing each dataset’s genre, provenance, applications, and size. Additionally, we also draft and present a tabular summary of the surveyed papers, emphasizing the different approaches, algorithms, and XAI techniques used for model interpretability, along with the modeling datasets and evaluation metrics applied.

3 Current Trends in Explainable AI for Fake News Detection

Building well-defined, explainable systems hinges on understanding their fundamentals. This notion is debated in the study by [33], which emphasizes the importance of a clear understanding of the fake news nomenclature, characteristics, and taxonomy. The authors answer the question of “What constitutes and qualifies as fake news” from a conceptual modeling perspective and advocate the relevance of this method for accurately defining, thus understanding, and communicating what fake news is, ultimately enabling and facilitating its detection. The proposed conceptual model hints on three crucial aspects of fake news namely the target, their opinion on a specific topic, and the extent to which the fake news will either strengthen or weaken that opinion. The fake news conceptual model comprises three sub-models:

- An attacker sub-model—which identifies the context from which the fake news originates.
- A target sub-model—responsible for assessing target population of the fake news, their opinion or reactive emotion to the fake news.
- A fact sub-model—specifies the relationship between true and false content and the extent to which facts are altered to produce the required psychological trigger.

The study provides some semantic insights into the identification of Fake news emphasizing the propensity for fake news being often built from distorted facts and the addition of reassuring content to make them seem more plausible. Another observed strategy for reinforcing the credibility of these fake news content is through the fake news authors’ persistence on authority figures either current or historical, or any other entity intricately linked with the subject of the Fake news. Moreover, fake news content is usually not rational but emotionally charged, to better manipulate people into credulously absorbing it. As a preliminary validation phase, the author tests the proposed approach on very small sample use cases, which produce promising outcomes, but which cannot be extrapolated for real-life use due to their limited size.

In line with this, [38]’s research analyzes the connection between fake news detection and emotion. The research proposes a Deep Learning-based interpretable text classifier named **TC-CNN (Textual Class-activation-mappings Convolutional Neural Network)** capable of assessing emotion, fakeness, bias, and levels of hatred in English news articles. The authors leverage the text classification powers of the **Class Activation Mapping (CAM)** technique, applying it to the pretrained FastText word embeddings created from the input data to explain the model’s outcomes. The CAM approach computes the contribution of each word per class modeled and then extracts the most relevant ones. The model is trained on 13000 “fake” English documents collected from the [Kaggle Fake News dataset](#) [39] and an equal number of “non-fake” entries randomly sampled from the [Signal News Media dataset](#) [40]. The efficacy of this initial task in predicting fake articles is evaluated, and the results are compared to those of a similar implementation proposed

by [41], whose architecture relies on **Recurrent Neural Network (RNN)** models trained with pretrained Glove word embeddings. Moreover, for a better comparison of these models, the authors also analyzed their results obtained with those from a 300d-Glove-based baseline model, the FastText classification framework [42] as well as a CNN-Prev model comprising a 3 layers convolution dense network of kernel size 3, and with a global max pooling architecture. Preliminary outcomes show the TC-CNN-Fake models outperforming all other models with Precision, Recall, and weighted F1 scores of 0.918, 0.919, and 0.918 while preserving a reasonable network size and thus better interpretability outcomes. Following [33] observation on the relationship between fake or misleading news and their heightened emotional contents, a second task is conducted by adapting and testing the proposed model on the **Text-emotion corpus** comprising 47 288 English tweets classified as either neutral, happy, angry, sad, or hate. For interpretability, the authors analyze the most frequent words extracted per predicted class by testing the method on a random selection of 1135 articles from 150 different US-based media, blogging, and news outlets. Then, a visual representation with text highlights of the most relevant features produced by the CAM approach's weight attribution techniques is presented, which helps infer bias and offenses from the news outlet data by outputting the learned vocabulary. Ultimately, the TC-CNN model outperforms all other classifiers, achieving Precision, Recall, and weighted F1 scores of 0.686, 0.688, and 0.683 for task 2, thus prompting the authors' conclusion emphasizing the importance of trustworthy news content used in the modeling process as a precursor for effective and human-understandable fake news detection systems.

3.1 Explainability Through Interpretable Features

The approach proposed in [45] involves utilizing interpretable features derived from two datasets: the **Information Security and Object Technology (ISOT)** research lab dataset [46], comprising more than 40,000 real and fake news articles, and the LIAR dataset [47], containing over 12,000 real-world short statements categorized as either "True", "Mostly True", "Half True", "Barely True", "False", and "Fake /Pants on Fire". The content from both datasets is analyzed for syntactic and semantic cues using CoCoGen [48], an automated computational and text analysis tool that calculates the weight and degree of coherence and cohesion within the text by utilizing a sliding window technique to capture the complexity of linguistic features. The method then calculates the local distribution of each feature score from within a text. For classification, the authors implement a modified version of the **Bidirectional Recurrent Neural Network (BRNN)** models with a **Gated Recurrent Unit (GRU)** [49], which considers and addresses the classes and metadata imbalances within the datasets during modeling. The most optimal subset of interpretable parameters retrieved is fed to the classifier through a feature ablation process using an adapted iterative sensitivity-based punning algorithm [50], which iteratively computes, ranks, and retains features with the highest sensitivity measures and reports outcomes at every step. The models' robustness is evaluated based on their overall global accuracy, precision, and recall, and the authors compare the model's outcome against those of SOTA black box RNN, CNN-based, and CAPSULE networks trained on text represented using word embeddings such as Glove, N-grams, and Word2vec. In the end, the results show that the proposed BRNN classifiers perform comparably to the black box models trained with sophisticated word embeddings, thus validating the authors' hypothesis that successful fake news detection can be achieved without sacrificing transparency.

The research presented in [51] investigates the discriminatory powers of approximately 172 features for detecting and classifying fake news. The study provides an overview of the impact and role of features in the overall decision-making process of fake news detectors by feeding a different combination of features to a fast and reliable Extreme Gradient boosting algorithm, XGB.

This algorithm can capture complex nonlinear interactions between features, which will be used to analyze the models' behaviors. To carry out the task, the researchers conducted a systematic survey to sample existing features from multiple fake news detection sources such as news content, reliable sources, and social media outlets. The features were implemented and evaluated on the Facebook [BUZZFace](#) datasets [52], which comprise 2263 factual contents extracted from articles related to the 2016 US presidential election. Entries in the dataset are classified as either "Mostly True", "mostly false", "Mixture of true and false," or "non-factual". For the experiment, the "non-factual" class was discarded while the combination of the "mostly false", and the "mixture of true and false" classes was used as the fake content, and the "mostly true" class was used as the true content. The impact of these features on the classifier was assessed by the construction of multiple unbiased models, each trained using a five-fold **Cross-Validation** (CV) approach, on randomly selected unique combinations of features. The predictive accuracy of the features was assessed by calculating the average standard **Area under the ROC Curve** (AUC) score for the models in which the features were included. Additionally, each feature's variability on the development set was measured using an average Mean Absolute Variation metric. The results reported show that only 5% of the features considered were included in models that scored an average AUC score higher than 0.85, while only 3% had relatively low variability. The relationship between model performance, the features used to train it, and AUC metric variability was further evaluated by analyzing the most common features in the best-performing models, considering results from models with the top 10% highest AUC scores and lowest variabilities. Ultimately, the authors create a cluster of features using the K-Mean algorithm to understand the prevalence and characteristics of features occurring in these models' spaces. Furthermore, the interpretability and explainability of the fake news detection model for each cluster are implemented using the SHAP XAI technique [53]. This technique captures the features most relevant classification features and uses that information to provide a visual representation of how different features impact each of the selected clusters' output decisions. The proposed method yielded some effective fake news detectors and was able to prove the importance of picking the right combinations of features in detecting specific types of fake news. Additionally, the author proposes some novel features for fake news detection, however, only a small fraction of the models trained achieve a detection AUC score greater than 85%, indicating the complexity and resource-intensive nature of the task.

3.2 Explainability Approaches in Combatting Covid-19 Misinformation

In 2020, the Covid-19 pandemic came to dominate world news. From global lockdown, shortages of everything, and businesses shutting down to rising death tolls, fear, anxiety, uncertainty, and insecurities were at their peak worldwide. Alongside the increased attention on the pandemic was a surge in Covid-19-related misinformation which wreaked havoc during the onset of the health crisis, creating confusion, exacerbating public distrust, and hampering effective health response. Proposed methods for countering the spread of misinformation ranged from human fact-checking [54], awareness campaigns (e.g., [WHO](#)), and a surge in related studies geared toward rapidly and effectively quelling this grave challenge. Appropriately, [55] proposes a framework for automatic fact-checking, fake news detection, and debunking, which involves building a database of verified sources from Covid-19-related entries collected from the [BBC News dataset for text classification](#) [56], as well as [PubMed](#) and [PMC](#), databases of health, biomedical and life science journals. The BBC news datasets were leveraged mainly for their non-medical macro-categories, namely entertainment, sports, tech, business, and political contexts, while the medical and scientific sub-macro-content were provided through articles retrieved from PubMed and PMC. Additionally, the [coronavirus news dataset](#), comprising 1.6 million news articles, was used as the main reliable reference for Covid-19-related facts, while the content of the [FNC-1 Dataset](#) was mainly used to

design and train the stance classification network used to retrieve useful features from text to assess what side of a debate an author fall on. The proposed framework works in four steps. The first and second steps involve matching the user's input query to the most relevant macro category and then pairing it with a specific sub-category of related content, such that queries that cannot find a match are discarded. This is done by leveraging the classification powers of four ML algorithms, namely **Support Vector Machines (SVM)**, **K-Nearest Neighbor (KNN)**, **Random Forest (RF)**, and **Naïve Bayes (NB)**, with the best model retrieved through voting majority ensemble methods. Subsequently, **Name Entity Recognition (NER)** with a **Bidirectional Encoder Representation from Transformers (BERT)** encoder and Multilayer Perceptron SoftMax decoder are implemented to help extract sentences that do not contain the relevant name entities from both the user inputs and the document. Then, a distributed memory model trained on the coronavirus news datasets with Doc2Vec embeddings is built, and the top K most relevant documents retrieved from the matched subcategory set are compared with the query via the Cosine Similarity method. In the last phase, the stance classification model network build is used to classify the topmost similar sets of documents based on the entry of query under four labels: agree, disagree, discuss, or unrelated. The stance model used comprised two input branches for the document and the query and was implemented on two model architectures: A **bidirectional Long- and Short-Term Memory (Bi-LSTM)** and a **Convolutional Neural Network (CNN)** architecture, which both achieved 65% and 78% accuracy; training for 50 and 700 epochs respectively. The method validation phase involves testing the models on new datasets- the **coVID19-FNIR [128]**, which contains true and fake COVID-19-related content. For this purpose, a cap on the number of similar documents is set and an analysis of the number of documents which "agreed" or "disagreed" with the query is done to determine either its authenticity or dubiousness. The results show an unreliable stance classification method that classifies most of the test set documents as unrelated, as such, is used for similar document retrieval and categorization.

In the same optics, [57] proposes a robust and dynamic two-stage Covid-19 fake news detection interface capable of estimating the veracity of an inputted claim, as well as providing a relevant explanation to the model's prediction confidence about the claim. The approach involves generating a collection of competitor explanations from a knowledge base of curated and verified information in real-time that is contextually identical to the input, and then matching, measuring, and analyzing the logical association between the claim and a fetched explanation from the set. To test the approach, the authors utilize a custom-made COVID dataset comprising 5500 false claims and explanation pairs sampled from the fake news debunking and fact-checking websites [Poynter.com](https://www.poynter.com). They used false claims from the same source and manually constructed true claims by rephrasing false claims, resulting in 1000 and 200 instances of cross-validated false claim-true claim and explanation pairs for training and testing, respectively. The 2-stage pipeline model architecture comprises component A, which is responsible for fetching relevant facts about the claim and involves training the transformer model using the claims and explanations as inputs and generating negative sets of pairs for the task. The candidate explanation built is then fed to the second component, which assesses the truthfulness of the claim by checking if its veracity matches that of candidate explanations retrieved from the dataset. Various contextual NLP techniques were implemented on both components for baselining, using the **Term frequency (TF)** and the **Term Frequency-Inverse Document Frequency (TF-IDF)** to represent and extract relevant features from the inputted text, and Glove embedding model to generate sentence vectors for the claim. Three transformer models, BERT, ALBERTa, and MobilBERT, were used for both components, characterized by their fast and real-time deployment ability and resource inexpensiveness. The pipeline's performance is evaluated based on component A's ability to extract useful information using the Mean Reciprocal Rank and Mean Recall @10 metrics and component B's expertise at

computing a claim's truthfulness based on explanations derived from the Mean standard deviation of Cosine similarity threshold between the tested claims and their explanation. Overall, the authors reported that the transformer models outperform the baseline NLP methods and that the combination of the best algorithms from both components A and B yields the best overall performance despite their high latency per claim and memory usage. However, the over-dependence of both models and the possibilities for errors running through the pipeline and affecting the outcomes constitute a significant drawback.

3.3 Explainability Methods for Mitigating Misinformation in Social and News Media

An innovative approach for Combating and attenuating the propagation of misinformation on news platforms is proposed in [58]. Here, the author designs a news reviewing, validating, and sharing interface that includes a built-in interpretable fake news detection assistant. The research aims to understand the impact of the detector's interpretability on the end user's confidence in sharing or sharing information produced by the system. Additionally, the study seeks to determine whether users would blindly trust the detector's outcomes even when the results do not make sense to them. To achieve this, experiments are conducted on datasets comprising news, rumors, and hoaxes from two sources: news story headlines and labels from the websites Snopes.com and the top 16 related articles crawled from the Google search engine. The datasets comprise 4638 and 30599 data points from each source, respectively. Four NLP classifiers are used to help generate diverse explanations for the tasks. In the end, the result from each algorithm is averaged, resulting in a final ensemble model with a 73.65% detection Accuracy. The algorithms implemented in this study include a BiLSTM network [59] with a self-attention layer- necessary to collect attention scores for each instance explanation and trained on Word2vec feature vectors derived only from news headlines. Additionally, there is a **Hierarchical Attention Network (HAN)** model [60] that computes attention weights to help select the most important sentences per article. The third approach used is a Knowledge Distillation model [61], comprising a BiLSTM teacher model trained on Glove embedding vectors derived from new stories, article contents, and their sources, and an XGB student model responsible for analyzing and ranking the most relevant feature importance in each iteration. Lastly, there is a BiLSTM model with an attention mechanism trained to get the most relevant features from news headlines and related articles using Word2vec embeddings. The fake news detector assistant enables users to report information as fake or skip it altogether, depending on their confidence level. It is also capable of providing the model's veracity prediction with or without explanation per the user's preference. This explainability is provided as visual elements ranging from heatmaps to Charts such as Bars and Pie, as well as an output list of the most important sentence features learned by the model. The effectiveness of the proposed AI framework is tested with a human-controlled user-subject study conducted following a between-subject design strategy with five conditions: These include a baseline condition supporting only human assessments, an AI-assisted condition with automated predictions from the ensemble model results (Without explanation), and three XAI assistant conditions, XAI Attention, XAI attribution, and XAI Attention+Attribution model. The study measures the effect each explanation has on the Human-AI collaboration by evaluating the participants' overall information review and sharing scores, their accuracy at guessing the models' outputs, and overall trust, reliance, and agreement with the model output on a scale of 0–100 %. The results show that the addition of an XAI assistant feedback with qualitative results not only improves users' overall systems evaluation performance but also helps them build appropriate mental models of the assistant in various scenarios, hence constantly adjusting the users' trust in the assistant accordingly. Nonetheless, the research reveals that most participants had a hard time learning the model's behavior and believed the assistant lacked common sense in distinguishing between fake and real news in certain cases.

In the same context, a triple framework of explainable fake news detection system called XFake is presented by [62]. The system is designed to assist users in confidently identifying fake news by analyzing news content from the perspective of data attributes, semantics, and linguistics. XFake was built using a dataset of 5104 fake news content, mainly comprising attributes such as subject, context, speaker, targeting, and statement, scraped from Politifact.com, a website of carefully curated and verified political news content. The XFake detector systems consist of three different Architectures: MIMIC, ATTN, and PERT, each trained on different aspects of the input targeting news and outputting a predictive score and supporting evidence. The MIMIC framework applies a **Neural Network (NN)** based teacher and **Decision Tree (DT)** ensemble student (80 decision trees) trained with Glove word embeddings, to study news attributes, hence providing explanation by outlining the key components and importance scores of the input news instance. The ATTN framework, on the other hand, analyses each news attribute through its phonological meaning. Its architecture comprises a pre-trained Word2Vec word embedding, a CNN model employed to provide local explanation, as well as an explanation based on n-grams analysis, and a Self-Attention mechanism for capturing global explanation and correlation between words. The last framework, PERT, is built to analyze input from a linguistics standpoint by extracting linguistics features from each input item and then feeding them to an XGB Classifier, which outputs the prediction outcome. The models' explainability is provided either as prediction scores and attribute importance visualized through histograms; visual highlights of the most relevant features using heatmaps, or through an outlay of the ensemble tree diagram showing cascading paths from the input to the prediction outcomes. The output prediction from each unit shows an accuracy of 67.1%, 67.3%, and 53.2% for the MIMIC, ATTN, and PERT frameworks, respectively. The overall system outputs a prediction as a unified weighted ensemble of these outputs to produce the system's prediction score corresponding to the news inputs' degree of fakeness. The system's effectiveness is tested on real-life scenarios through a human study involving 147 Amazon Turk Workers fact-checking and guessing the prediction of each claim to test the adequacy of the AI-generated explanation, as well as the user's perception of the system. Conclusively, the study proves that explanation is crucial for humans to better comprehend and predict system actions. However, users thought the addition of explanation, contributed to slowing down the humans' assessment of the systems since it took them more time to study, interpret, and understand the detection results.

Detecting fake news on social media is a challenging task due to the massive scale of user-generated and anonymous data, which is often styled specifically to deceive readers and evade detection. User comments can provide important clues to assert the veracity or fakeness of the content. This premise is developed in [63], which proposes an explainable fake news detector called dEFEND designed from social media data. The approach aims to extract explainable features from noisy data sources by leveraging both the news content and users' stances, opinions, or comments on the online platform. This is done such that the most relevant features collected positively impact the detector's performance. The framework comprises four components namely, a Hierarchical Attention Neural Network based news encoding component, which captures relevant context and interpretation of the news content using **Bidirectional-GRU (Bi-GRU)** to capture contextual cues and dependencies at the word and sentence levels. The user comments encoding component also leverages the Bi-GRU to learn the latent representation of users' comments. The third component is the sentence-comment co-attention level, which analyses the relationship between the users' comments and news content, collecting important correlated features necessary to assess the news' degree of fakeness. The outcomes produced are then integrated into the fake news detection to predict the veracity of claims. The proposed system's effectiveness is tested on the [FakeNewsNet dataset](#) [64], which is collected from fact-checking platforms such as GossipCop and PolitiFact. The dataset comprised labeled news content, users' comments, and engagement

information while ensuring only articles with three or more comments were sampled. The detector's efficacy is validated by comparing its performance against Seven SOTA fake news detectors, namely, the **Hierarchical Attention Neural Network (HAN)** [60], the **HPA-BLSTM** [65], the **Rhetorical Structure Theory (RST)**, the **Linguistics Inquiry and Word Count (LIWC)**, the **text-CNN** [66], the **TCNN-URG** [43], and the **CSI** [44]. The models' performance is evaluated based on their Precision, Recall, F1 score, and Accuracy, and the results reported show dEFEND consistently outperforming similar methods based on news contents and user comments on both datasets.

A follow-up analysis to test the impact of news content and user comments on the detector's performance involves designing and implementing three variants of the dEFEND tool, i.e., one that does not consider user comments, another one void of news contents, and one lacking the sentence-comment co-attention components. The results from which authors assert that users' comments and news content play an important role in improving the fake news detection capability of the method, as they are interdependent and hence necessary for the task. To evaluate dEFEND's explainability, the HAN and HPA-BLSTM models' attention mechanisms are leveraged as baselines to assess the news contents and the user's comments' explainability, respectively. For the news sentence explainability, the author leverages the ClaimBuster corpus [67] to create a ground truth rank list of check-worthy sentences for all inputted news content. This will be useful to determine whether the top-ranked explainable sentences proposed by dEFEND are check-worthy. In this case, a sentence is presumed to contain check-worthy claims if it is assigned a high corresponding score from ClaimBuster. On the other hand, the users' comment explainability for each article is evaluated by assessing the explainability ranks list for the top K comments outputted by dEFEND and HPA-BLSTM. This task is undertaken by Amazon M-Turk Workers, who are tasked with reviewing both lists and providing a general list-wise comparative assessment and a more specific item-wise evaluation of each list's comments. From these studies, the dEFEND model tends to excel in both tasks and proves best at ranking more explainable comments and at spotting the most relevant explainable features from comments. However, it relies heavily on a community of dedicated readers, and fact-checkers, which may not always be readily available.

A surrogate investigative study into the interpretability of SOTA fake new detection tools, such as dEFEND, HDSF [68], and TCNN-URG, is proposed by [69]. The author provides an assessment of the global and local features that contribute to explaining the models' decisions. The particularity of the detection models compared in this study is that they are modeled from news content and users' comments. While the HDSF model focuses more on the structure of the news content and the TCNN-URG on both news content and users' comments, dEFEND utilizes users' posts and comments or opinions to spot fakeness. Initially, the efficacy of each model is evaluated based on Precision, Recall, Accuracy, and F1 Scores on PolitiFact's [FakeNewsNet dataset](#), which comprises only articles with two or more comments. The preliminary results of testing each model on this dataset show that dEFEND and TCNN-URG were quite adept at capturing both syntactic and semantics cues from input data containing both users' posts and comments. However, dEFEND performs best with an Accuracy, Recall, and F1 score of 0.82, 0.81, and 0.80, respectively, while the latter model achieved the best precision score equals 0.80. The models' explainability is assessed using three model-agnostics explainers, namely Captum [70], SHAP, and LIME [71], leveraging LIME mainly for local explainability and SHAP and Captum for global explainability by analyzing the average SHAP values for the top 100 important features extracted using CAPTUM. This global assessment helps evaluate and create a mapping between the prediction gaps for the positive and negative classes and a model's trustworthiness. This approach results in the model's capability to select the features that most inform its correct prediction outcomes, even though most of the top explainable features outputted contain quite a few meaningless tokens.

In a follow-up analysis, the authors select the top 100 positively predicted comments, stripped of these useless words, and report a significant boost in the overall quality of the interpretable feature extracted, with dEFEND outperforming all other detectors with a 7.5% boost when compared to the second-best interpretable detector, HDSF. For local explainability using LIME, the authors evaluate and highlight the top 15 features extracted from a positively predicted random selection of fake and real news content from each detector. TCNN-URG stood out as the detector with the most explainable outcomes, while HDSF performed the worst, and dEFEND performed averagely but captured fewer useless words. The authors then attempt to analyze the model's prediction capabilities based on their false positive predictions, using LIME to emphasize the impact of each feature on the detector's outcome. For this task, HDSF did not show any explanation outcomes, while TCNN-URG explained the article better than dEFEND. A human subjective test is also conducted, which involves 36 end-users tasked with evaluating, on a 5-point scale, the trustworthiness of a random selection of 30 LIME-explained fake posts correctly predicted by all three models. In the end, the TCNN-URG model performed best with an average score of 3.5, compared to 3.05 and 2.5 for dEFEND and HDSF, respectively, despite dEFEND achieving an outstanding accuracy score during modeling. The results prove that model explainability impacts human trust in a system more than the model's overall prediction performance. They also emphasize the importance of effective feature selection and the right interpretable/Explainable model choices.

3.4 Explainability Methods for Neural Nets, and Attention-Based Misinformation Detectors

In recent years, NNs and Transformer models, with their sophisticated attention mechanisms, have significantly advanced the field of AI, Computer vision, and NLP. These impressive ML models have shown exceptional ability to understand and analyze natural language, thereby improving the accuracy and nuance in identifying harmful and misleading online content. Deep learning architectures within NN are adept at detecting complex patterns in textual data, while transformer models like BERT and GPT utilize attention mechanisms to emphasize the most pertinent parts of the input, greatly enhancing contextual understanding. Despite their impressive capabilities, these models face criticism for their "black box" nature and the substantial requirements for labeled data, computational resources (typically GPU accelerators), lengthy training times, and high power and energy consumption according to Angelov et Soares [24]. However, [72] presents an analysis of the explainability of fake news detectors built from SOTA Neural Net (NN) and Transformer models and argues that the accuracy and explainability of these models are greatly influenced by factors such as preprocessing operations, text length, and complexity. The study evaluated the effectiveness and transparency of a BiLSTM CNN model with attention mechanism and a transformer model, BERT, in accurately detecting fake news. The experiment is conducted on the [FakeNewsCorpus datasets](#), which comprises approximately 9 million texts classified as false, satire, bias, conspiracy, state, junk science, hate, clickbait, unreliable, political, and true. For this study, only the content from the "Sports" and "Tech" subsections are used, with a balanced number of 1000 randomly selected articles from each subsection, each set containing an equal number of 500 fake and 500 real articles. The study also investigates the impact of various preprocessing techniques on models' performance by replicating 16 additional sets, each comprising 1000 articles, resulting in a total of 9 sets from the "Sports" and "Tech" classes, respectively. Each of these sets is then subject to varied levels of text preprocessing methods ranging from a vanilla class with No preprocessing applied to classes with maximum article summarization lengths of 5 and 10 each, accounting for word TF provided by the TF-IDF method and ensuring the removal of stop words, resulting in a 68% summary accuracy. Additionally, text stemming and lemmatization methods are applied to every other class. Both algorithms are fed with the text created, and their

classification effectiveness is evaluated based on their overall accuracy and attention coverage. Preliminary results show that the NN-based models achieved their highest accuracy of 85% with the vanilla class compared to only 52% accuracy for the BERT model. During the prediction phase, attention weights are produced, which help visualize the networks' interest in each processed word and provide insight into the models' classification decisions. This also sheds light on the underperformance of the different summarization approaches, attributed to the methods' inability to narrow down and emphasize relevant contextual cues necessary for the task, most of which may have been lost during the text summarization phase. However, an investigation into the model's attention focus for the summarized texts reveals that BiLSTM-CNN has almost double the coverage of strong attention compared to BERT, although the BERT model is more selective in assigning attention to words and has twice the attention coverage for weak attention. Moreover, both models consider nouns and verbs as the primary and most influential parts of speech in their model classification phase, and they are also effective at spotting melodramatics and hyperbolic semantic language patterns known to be highly noticeable characteristics of fake news content. Overall, this research result confirms the superiority of deep NN models over BERT, particularly when applied to short texts. They also emphasize the adverse effects of certain NLP text preprocessing techniques on the model's performance, as these techniques may alter crucial semantics and contextual cues within the text. A drawback to the approach is the use of accuracy as the only model evaluation technique, as it has been shown to produce unreliable outcomes, especially when data is unbalanced.

An alternative NN-based method fake tweet detection method called GCAN-Graph-aware Co-Attention Network, which relies on a dual co-attention mechanism, is presented in [73]. The proposed approach aims to analyze the extent to which users' features extracted from their profiles, tweets, and retweet patterns could be leveraged to infer content trustworthiness, as well as provide a visual highlight of relevant features that contribute to the detector's outcome. The dual co-attention mechanism aims to understand whether there exists an underlying relationship between the source tweet content and the user's retweet patterns. The GCAN comprises five components: a user's online content/ interaction extraction phase, a second phase that encodes source tweets extracted, a user propagation representation phase powered by GRU and CNN algorithms to help create a roadmap for understanding tweet propagation between users, and the last phase that employs a dual co-attention mechanism implemented through Graph Convolution Network. This phase is responsible for generating feature vectors between source tweets and the users' tweeting patterns and identifying a correlation or mutual influence between them. In the end, the co-attention vectors and learned representation produced are aggregated and used to determine a claim's veracity. The detection model is tested on the Twitter15 and Twitter16 datasets [74], which comprise 190 868 and 115 036 source tweets and corresponding collections of users' retweets, respectively. Only instances labeled "True" and "Fake" are considered ground truth, and the users' characteristics and features needed for the tests are retrieved from the user's ID information by crawling the X (formerly known as Twitter) API. The efficacy of the proposed framework is validated by comparing its performance output against those of eight fake news detection models namely DTC [75], SVM-TS [76], MGRU [77], RFC [78], CSI, TI-CNN [79], CRNN [80], and dE-FEND, based on systems' Accuracy, Precision, Recall, and F1 scores. Additionally, a variant of the model GCAN-G, which lacks the graph convolution part, is created to compare with the original methods and assess the importance of the user's tweets to retweet patterns or vice versa on the models' efficacy. From the result reported, GCAN outperforms all other detection methods, on both datasets, with an overall average performance boost of 17% and 15% for the Twitter15 and Twitter16 datasets, respectively, compared to only a 14% and 13% boost with the GCAN-G algorithm. This underscores the importance of the dual co-attention mechanism for such tasks. Another

important revelation is the fact that a dual co-attention framework could be used to curtail early misinformation spread, as this method also proved quite effective at that, with a 90% accuracy at detecting fakeness after only 10 retweets. The model's explainability is assessed by studying the distribution of co-attention weights implemented on the source tweet word, the retweet propagation patterns, and the retweeters' characteristics, and then visualizing these relationships as text highlights or on Heatmaps. The outcome of this task corroborates [72]'s conclusion that fake news contents often tend to be quite dramatic and doubtful and insists on the importance of analyzing early retweeter behaviors in assessing the veracity of a story. Furthermore, the study reveals patterns such as unverified and newly created accounts, the brevity of user description, and their closeness to the original news poster as potential retweeter characteristics to consider when trying to analyze and identify fake news spreaders.

Relatedly, [81] proposed a method that also considers users' profile information, tweet contents, and retweet patterns to build their fake news detector - **Multi-View Attention Network (MVAN)**. However, their Deep learning approach relies on a multi-attention component called **Graph Attention Networks (GATs)**, for news propagation encoding and representation. The proposed mechanism comprises three components: a text semantics attention network that leverages Word2Vec to encode the source input, and a deep **Bi-directional Gated Recurrent Unit (BiGRU)** to correctly represent key cues from the source tweet text. Then, there is a propagation structure attention network inspired by Graph Attention (GAT) networks for capturing helpful insights relevant to the tweets' propagation structure. The last component uses all the information retrieved to predict a tweet's trustworthiness. The MVAN tool is tested on the Twitter15 and Twitter16 datasets, where the user's original unique identifiers are leveraged to extract their characteristics by scraping X. This results in 77,049 and 108,801 user feature information retrieved for each dataset, respectively. The detector is built on preprocessed text with Word2Vec embeddings, and its fitness is assessed by comparing its performance against that of eight other methods namely, the SVM-BOW [74], BiLSTM, TextCNN [82], CSI, CRNN, dFEND, GCAN. The reported results show that MVAN outperforms all other models based on their Accuracy, Precision, Recall, and F1 score, also achieving an early fake news detection accuracy score of 91%. In a follow-up ablation analysis of the models to determine the relevance of each component, it is reported that the semantics and propagation structure attention mechanisms play a crucial role in enhancing the tool's performance. The model's interpretability is assessed by analyzing the word's semantics and propagation attention weights values to capture important clues used by retweeters and then visualizing the most relevant word that impacts the model's decision outcome through text highlights. The overall observation reported tends to agree with [73]'s conclusion about certain special revealing characteristics of fake news spreaders, including old and overdue accounts, fake profiles with invalidated account icons, little/no followers, and following patterns.

Despite the effectiveness of these attention mechanisms at spotting fake news and visualizing explainable features, researchers [83] have expressed skepticism about the continuous reliance on attention mechanisms for models' explanations. Some have gone further to propose alternative transformer-based applications, such as exBERT and VisBERT, that can be leveraged to assist users in understanding BERT models' decision-making processes. For instance, VisBERT can be used to visualize contextual cues within the BERT models question-answering task and works by focusing on the hidden states output for each encoder block to analyze the semantic representation transformation throughout the modeling process.

3.5 Explainability as an Add-on Component to Misinformation Detectors

A framework providing an explainability component as an add-on to existing BERT-based fake news detection models is proposed by [84]. In their implementation, the original structure of the

detector remains unaltered, but a model-agnostic XAI-based module, either a LIME or ANCHOR [85], is affixed to the tool to facilitate its interpretability. The proposed experimentation is built from a BiLSTM and DistilBERT-based fake news detector, respectively trained on a sample of 10,361 real and 7,924 fake instances collected from the [Fake news dataset](#), and 21,471 verified and 23,481 fake news samples collected from [the fake and real news datasets](#), respectively [86–87]. The experiment is conducted in three phases. The retrieved datasets are formatted to resemble social media content posts. The data is then modeled on the classifiers to distinguish between real and fake content. The models' performance is evaluated on their respective datasets based on overall Accuracy, Precision, Recall, and F1 scores. The initial classification results show that DistilBERT slightly outperforms the RNN-based classifier across all metrics and classes. An XAI surrogate-type tool is configured in the next phase, by investigating the architectures' interpretability. This involves assessing each of the models explained with LIME based on their classes' prediction probability outcomes, the most relevant word impacting the model's decision, and their calculated weights. The models' explainability evaluation with the Anchor method is also evaluated based on their outputted precision scores for particularly influential Anchors that defined the sentence's veracity. In the end, an easy, convenient, and explainable add-on module is built that can be adapted to a plethora of fake news detection models to provide more transparency and insight into the model's classification decisions. The results reported that the overall model predictions and explainability outcomes may be highly influenced by certain name entities and settings, often highly associated with fake news. However, the effectiveness of the method at providing accurate explanation relies on using multiple surrogate-types XAI techniques, and the Anchor-based explainability approach's failure in certain scenarios constitutes a significant drawback to this method.

A data model and architecture based on Linked Credibility Reviews, which can be incorporated into distributed bots designed for automated fact-checking and fake news detection, is presented in [88]. This proposed implementation works by reviewing and assigning a credibility score to a news source or content, while keeping tabs on the information provenance such that the system's explainability is provided by human retracing the initial content writer and fact checker or reviewer ground truths and assessing the accuracy of the scores initially provided for each content. The credibility system aims to provide a human's perspective on how much the credibility ratings assigned to a claim by the bot corroborate or refute the input data and how those ratings compare to those subjectively assigned by humans. The presented NN-based bot framework – *acred*, includes two **Ground Credibility Signal (GCS)** lookup Bot, a linking bot, and a decomposition bot. The GCS bots generate a query based on the item, either a claim or a website under review, and assign to it a credibility rating based on relevant information about the claim extracted from the relevant and corresponding GCS source. The GCS sources comprise a database of ClaimReviews, which provides accuracy ratings for claims from human fact-checkers, and a tool that provides accuracy ratings for websites, either through expert or crowdsourcing rating. The linking bot helps generate CR for documents by dissecting, analyzing, and extracting the sentence(s) with the most relevant factual cues. Then, a credibility rating is assigned to the new input sentences by mapping them to one that already has a credibility score. The decomposition bot is better suited for identifying relevant content from Creative works such as social media posts, podcasts, online articles, and so on, and assigning CR to the most relevant parts of the content being reviewed from such venues. The efficiency of the proposed fact-checking system architectures is validated on datasets comprising a mix of social media posts, political speeches, and news articles. The three data sources users are [The Clef'18 Check! That dataset](#) [89], consisting of claims to be categorized as either True, Semi-True, or False; a sample of 420 fake and 528 real texts extracted from the [FakeNewsNet](#), and 250 annotated tweets from the [Co-inform250 dataset](#). The prediction procedure for each data input involves reading and formatting the input content to the appropriate style, requesting a credibility

rating from the acredited bot, then comparing the provided review against the initial content labels, and saving them for later analysis. Preliminary results show that despite the SOTA performance achieved with the Clef'18 datasets, the models were biased and overly confident on a certain task. Therefore, the original architecture's component structure is updated with optimized sets of parameters and re-evaluated on the three datasets. The models' performances on both architectures are measured and compared based on their confusion matrix outputs, the Macro Mean and Mean Accuracy Error, Accuracy, Macro F1 score, and Macro Average Recall for the Clef'18 dataset; Accuracy, Precision, Recall, and F1 score for the FakeNewsNet dataset; and models accuracy on the Co-Inform250 dataset. In the end, the revised framework- Clef+ performs even better across the board, even though the architectures show poor precision scores for real or unverified data, and in some scenarios, the explanation produced does not match the systems prediction label.

4 Current Trends in Explainable AI for Hate Speech Detection

Several instances in history confirm that misinformation and hate speech have been the main precursors for Mass atrocities, grave human rights violations, and even Genocides. The Holocaust, Rwandan, Cambodian, and Srebrenica genocides, and most recently, the Myanmar crises did not start with the mass killings but with dominant regimes implementing a series of misinformation and hate speech dissemination campaigns against the minority groups. Nowadays, the increasing adoption of social media through reshares, clicks, and likes is contributing to exacerbating this problem which has already claimed multiple instances of Hate crimes [90–91], and domestic terrorism [37, 92]. A recent study by [93], involving data collected from 163 countries spanning 17 years, shows a direct relationship between online misinformation and political hate speech and how they contribute to accentuating domestic terrorism in these countries. The problem of hate speech is grave, and the urgency to detect, predict, and ultimately, reduce it online has led to an uptick of research studies implementing ML algorithms for this purpose, as well as an increasing demand for more accountable and trustworthy hate speech detection systems that are transparent and easily interpretable to human. In this section, we provide a collection of the most recent bodies of literature leveraging XAI techniques applied to detecting offensive and hate-related languages, contributing to the creation of transparent automatic detectors for social hate and insensitivities.

4.1 Explainability Through Interpretable Features

From a Deep NN viewpoint, the authors of [94] investigate the explainability of Deep Learning-based hate detectors by leveraging a computer vision technique called partial occlusion. This technique is used to visualize and help localize sensitivity spots within the input data. An adapted version of the image classification method was applied to a CNN-GRU classification model trained on the English offensive language dataset [95] to identify specific words or phrases in the input data that could be vulnerable to missing information. Partial occlusion was used to evaluate the robustness of the classifier to missing information and works by iteratively blocking parts of the input sequence with a token and monitoring how the model adjusted to these changes. The classifier's outcomes are analyzed throughout the process, with their outcomes displayed as heatmaps, highlighting the computed sensitivity probability for each word with respect to the true class. The maximum activation method proposed by [96] was implemented to understand the most learned and predictive features. This involves extracting the top input features that maximally stimulate individual neurons by computing the model's unit activation scores for each of the input tokens. Each neuron is further analyzed, and results show that the CNN-GRU is effective at spotting interpretable lexical and syntactical cues, even though, the method's final activation layer often fails to capture contextual semantic distinctions between tokens and usually produces vague or

non-interpretable outcomes. A follow-up experiment is carried out to identify and extract the most significant hate speech unigrams feature. This is done by feeding the CNN-GRU model with semantically neutral test examples composed by filtering out unique words and non-alphabetical tokens across tweets and manually constructing the input token such that each of the synthesized examples contains a noun and an adjective. This resulted in 101 top unique features grouped into nine distinct classes, categorized as either containing hateful content or not. The quality of the synthesized tweets from the group with late terms was determined by evaluating and comparing the recall score computed against that of a random baseline recall of [Hatebase](#) lexicon terms. The proposed method achieved a better recall score than the baseline models, and it was also reported that synthesized features in plural form also tended to be most accurately predicted, hinting at the possibility that hate speech detection models could be better suited if implemented on the text describing communities rather than in an individualistic setting. However, a downside to this approach is the model's decreased representational capabilities on lengthy input sequences and their high error rates due to the models' lack of class separability.

In a similar method proposed by [97], a focus is placed on building an explainable detection model for hate speech and toxicity in the context of Covid-19. This is achieved by utilizing important features extracted through XAI methods and applying NLP text classification techniques to help the models output more interpretable outcomes. The approach leverages prior content extracted from the model training step and uses them in the classification phase to help reduce false positive rates and maximize the detector's performance. The author utilizes contents from three datasets of hate and offensive language collected from the **Offensive Language Identification Dataset (OLID)** [98], the HateEval – Multilingual Detection of Hate Speech Against Immigrants and Women in X [99], and the racist and sexist tweet dataset [100] in their training and evaluation phase. The merged contents are subcategorized into three classes, emphasizing whether the offense is directed toward immigrants, women, individual subjects, and/or political content. To test the generalizability of the proposed method on Covid-19-related hate content, the **Texas Advanced Computing Center (TAAC)** Covid-19 X hate speech dataset19F, and the Georgia State University Panacea Lab [101] dataset of offensive racial content is used. The text classification baseline model implemented follows an existing approach by [102] and utilizes Glove embeddings for tweet transformation. The data is modeled using a CNN classifier, and the optimal output parameters are selected by sorting the validation AUC scores computed. The proposed method works by calculating the global average feature importance for each selected term per class using the SHAP's Gradient Explainer methods and the local feature importance for each observation predicted by the model. The percentage difference between them is then computed, resulting in an optimal predicted probability score. The effectiveness of these enhanced prediction methods is then tested on the COVID-19 data, which shows great results at correcting false positive classification and a propensity for attributing contents to the negative class. However, the models tend to produce conflicting classification classes for certain scenarios, notably tweet content directed toward specific ethnic groups or migrants.

4.2 Explainability Through a Human-in-the-Loop (HITL) Mechanism

The ERASER benchmark - Evaluating Rationales and Simple English Reasoning proposed by [103] is a novel approach to model interpretability, which comprises multiple manually annotated datasets and tasks, with supporting evidence and information known as "Rationales" relative to their annotator decisions regarding a claim. This benchmark presents two measures: Plausibility and Faithfulness, as ways of revealing the model's decision-making process by measuring how much the models can be impacted by the erasure of either the most or least important features. The plausibility metric is the measure of how convincing a particular interpretation is to humans,

while the faithfulness metric, subcategorized as comprehensiveness and sufficiency, is a measure of how much an important token truly contributes to a model's reasoning process. Comprehensiveness assumes a model is faithful if the most relevant tokens are widely representative of the whole input text, while sufficiency measures whether the important feature contains sufficient information to retain the original prediction. The ERASER approach works by evaluating how well the rationale predicted by an NLP model matches those provided by human annotators and the extent to which these human rationales impact the corresponding models' predictions. Models that apply the ERASER benchmark or consider incorporating human rationales in their quest for more interpretability have been lauded for their efficiency and ability to produce outstanding performance results and reduce unintended bias often directed toward certain marginalized communities [104–105]. The concept of incorporating human rationales into the models' interpretability investigation has also contributed to correcting a common misconception about an if-then relationship between the model's performance and its interpretability, especially in terms of plausibility and faithfulness metrics [104, 106]. The idea of using rationales was introduced by [107] and presented as a way of getting human annotators to highlight a specific section of a text that support their labeling decision of the claim, such that those same rationales guiding a model decision would contribute to making the models more human-like and hence more understandable and interpretable.

Mathew et al.'s [104] work heavily relies on the use of rationales and their implementation in modeling to improve hate speech detectors' explainability. Their experiment relies on a large collection of 20K hate speech instances – HateXplained- extracted from X and Gab.com, considering only posts categorized as either hate, offensive, or normal. The content of the HateXplained12F dataset is annotated by Amazon M-Turk Workers, who, in addition to producing claim labels, are also required to provide the target community with supporting evidence to which they believe the hate or offense is or might be directed. The 919 class labels for which the annotators had dissimilar opinions were discarded. The class labels and the target community remaining were validated through a Majority Voting approach, thus resulting in 10 targets, namely, the African, Islamic, Jewish, Gay, Women, Refugee, Arabic, Caucasian, Hispanic, and Asian communities. On the other hand, the rationales collected from the hate or offensive posts are converted into attention vectors and utilized to generate ground truth attention for the task. The efficacy of the proposed HateXplained benchmark dataset is evaluated on two fronts, either by training a model using the class labels and ground truth attention vectors generated or exclusively on the original ground truth class labels i.e., Hate, offensive, or normal. The models used for experimentation include an adapted version of the CNN-GRU model [108], comprising a one-D filter with extra window sizes of 100 filters each, the BiRNN-attention model, and the BiRNN [109] models fine-tuned with LSTM and GRU, and the BERT-base uncased model. It is worth noting that the BiRNN and CNN-GRU were used only in the latter scenario, as labels from this set were not adapted for the outputting vector that represents attention. Additionally, the authors utilize the common crawl version of the pretrained Glove model to build the word embeddings for the non-BERT classifiers, and each model's classification performance was evaluated, compared, and reported in terms of their accuracy, Macro F1 score, AND AUROC scores. A bias analysis was undertaken to determine the hate speech detectors' propensity for making biased predictions against victim groups, thus helping assess the models' ability to avert unintended bias. This was measured using an AUC-based metric adapted from [110]. Ultimately, the authors attributed BERT's outstanding performance and aptitude for mitigating unintended model bias to the model's capability of seamlessly incorporating human rationales into the modeling process, even though the same cannot be said about their explainability. LIME was the explainability XAI method implemented, and the best tokens for computing the model's explainability metrics leveraging the ERASER benchmark were selected by considering modeling with attention or with LIME. For that purpose, in an investigation into the models'

overall explainability following the ERASER benchmark's plausibility and faithfulness metrics, it was observed that modeling with LIME turned out to provide more faithful results comparatively. However, the best-performing classifier and most effective bias-reducing model did not perform as well on the ERASER benchmark metrics, thus validating the observation that a model's performance alone does not suffice in assessing its effectiveness and trustworthiness.

Correspondingly, Mehta and Passi [105] investigate the ability and decision-making process of SOTA ML and transformer algorithms to detect and categorize social media hate speech, using LIME for Interpretability and the ERASER benchmark to assess the models' subjective explainability. The authors trained BERT, an Artificial NN, A Multi-Layer Perceptron, a DT, a KNN, a **Multinomial Naïve Bayes (MNB)**, a RF, LR, and LSTM on the Google jigsaw dataset, consisting of around 300,000 instances of utterances classified as a clean, toxic, insult, identity hate, severe toxic. The classification performance of each model was evaluated based on their Accuracy, Precision, Recall, Specificity, and Macro F1 score, with the LSTM, MNB, and LR models achieving the best accuracy. RF and KNN outperformed in terms of Precision, while the DT model scored best in specificity. The LIME method was applied to each of the linear non-complex models, namely, RF, NB, DT, and LR, and the results show an overall explainability hate speech prediction probability of 90%, 80%, 100%, and 95%, respectively. The authors then trained the HateExplained dataset [104] on these LIME-Explained linear models and on an adapted BERT model, whose architecture was modified to incorporate the MLP and the ANN models, to produce robust variants of the BERT models, namely- BERT+MLP and BERT+ANN. The interpretability of the BERT variants was provided using a layer-wise propagation approach to help identify the most or least relevant features in the decision-making process. In the end, the BERT+MLP and BERT+ANN outperformed the linear LIME-explainable models with accuracies of 93.67% and 93.55%, respectively. The models' overall explainability was then assessed on the ERASER benchmark by computing the models' plausibility scores based on the intersection over union metric, the Token F1 score, and the Area under the precision-recall curve metrics, as well as the models' faithfulness by measuring their comprehensiveness and sufficiency scores. The result from this analysis shows that the BERT variants provided the best and most convincing interpretations to humans, with the BERT+ANN being the most comprehensible based on subjective rating by humans, owing to the model's simpler structure and architecture. A follow-up analysis into the models' unintended bias tendencies also confirmed the superiority of the BERT variants models as the most unbiased model at categorizing hate speeches as either toxic or nontoxic and the best model for reducing unintended model bias. The authors' conclusions tend to reiterate the aptitude of BERT models to produce outstanding model performances, providing the most convincing interpretability to humans and helping mitigate the model's unintended biases.

4.3 Explainability Methods for Combatting Political Bias and Mitigating Hate in Online Social Groups

In a study investigating the impact of politically biased news data on the classification performance of a hate speech detector, the authors of [111] assert the negative effects of politically biased data on hate speech detection. They demonstrate how XAI techniques could be leveraged to visualize and anticipate such models' biases. The proposed approach experiments on three politically biased datasets: the political bias hate speech corpus⁹ built from the GermeVAL and GermeVAL task2 datasets, which comprise a binary class of offensive and non-offensive tweets. Only the offensive instance from these datasets is collected and combined with non-offensive content originating from politically left-wing, politically right-wing, and politically neutral sources extracted from the X API. To ensure dataset uniformity and comparability of topics with the original non-offensive records replaced, the **Latent Dirichlet Allocation (LDA)** algorithm is used to build a

topic model. The normalized pointwise mutual information optimizer is then used to find the most optimal number of topics with 10 keywords each. The non-offensive tweets collected are scrutinized based on their topic probability scores, and only those with the highest scores are considered for the experiment. In the modeling phase, a hate speech classifier is trained following a three-fold CV approach using a data sampling method where a proportion of the non-offensive class from the original political dataset -GermEval, is iteratively replaced by each of the three politically non-offensive extracted data on a 33%, 66%, and 100 % ratio for each respective subnetwork. A BiLSTM model is used to study the effects of these political biases, and its classification performance is assessed based on the output F1 scores. For explainability, the SHAP values for each trained token are computed to help spot and highlight the most important features impacting the model's decision. The preliminary results reported tend to disagree with the authors' initial hypothesis, which asserted equal models' performances across the board, irrespective of the size of the offensive data substitution and replacement. The model's output reported shows them producing different F1 scores, i.e., 84.8%, 83.1%, and 78.7% for the politically neutral, left-wing, and right-wing subnetworks, respectively. It can be observed that the politically neutral datasets consistently outperform the other subnetworks with growing replacement rates, hence proving that politically biased data in the training process negatively affects hate speech detection. After a series of analyses, the authors conclude that the degree of model impairments depends on the dataset's political orientation bias, following from the results that politically right-wing data affects the classifier's performance the most. However, criticism of this work points to the high disconnect between the synthesis of politically biased data simulated and used in the experiment with the real-life data applied. Another limitation comes from the inconsistency and the time of data retrieval periods for the samples used in the experiments.

In another study, the authors [112] investigated the feasibility and impact of users' past tweets and network data on the performance of a multimodal abusive language classifier, as well as how XAI techniques could be leveraged to make the model's outcome trustworthy and interpretable to humans. The three datasets used for the task comprised 14,939 English tweets classified as either hate, offensive, or neither, extracted from [95], along with the user content and data network retrieved from the X API. The second dataset has 16,907 English tweets grouped as either sexist, racist, or none, collected from [100] and including user and network information. The third dataset was provided by [111] and contained a sample of 68,443 German offensive/non-offensive language instances with network and user information, labeled manually, and automatically using the BERT classifier, with the small sample of manually labeled instances utilized as a test set for the experiment. For the classification task, the authors utilized the pretrained DistilBERT model with classification head, with the top 500-word tokens from the user's tweet history vectorized using the TF-IDF BOW method. An unsupervised inductive learning framework – GraphSAGE, particularly useful in generating embeddings for new unseen data, was employed to model the users' network data by modeling the graphs such that users' nodes are derived from their tweets annotations and assigning a zero to users with no network connections. The classification input was retrieved from a combination of the classifiers' outputs, which were then fed into a linear prediction layer with SoftMax activation to produce the desired number of classes. Due to data imbalance, data sampling techniques were applied during the training phase, and each sub-model was trained independently. The model's prediction capability was assessed based on the outputted Precision, Recall, and F1 scores, and the results indicate that the model with the best classification F1 score contains all three data components, i.e., the user's text, history, and network information. It was also observed that the proportional increase in F1 score for each model relative to their network's density might indicate that the user's content and social network information do indeed positively affect the abusive language classification performance. To achieve model explainability,

Shapley values were computed across two granularity feature levels i.e., either component-wise or feature-wise. These scores are calculated by iteratively averaging the features' impact score on the models' prediction outcome, and from these results, it was noticed that the text model plays a crucial role in the classification outcomes, followed by their tweet history and then the network information. Also, the output results show a positive correlation between the user's tweet history size and Shapley values, hence validating the author's assertion about the importance of user networks and history in abusive language classification. From this, a follow-up analysis on the model's feature-wise explainability was carried out to help highlight and visualize each token's overall contribution within a sentence. These results obtained tend to confirm the author's second hypothesis that abusive model explainability can be effectively achieved at a global level, as well as a single feature prediction level.

An ULMFIT-based Explainable hate speech detection system adapted for online social group moderation is proposed in [113] through a design science two-step research implementation approach borrowed from [114]. The author leveraged this framework as a guideline template for creating a social media dashboard interface into which a hate speech detector would be integrated to help spot uncivil language during interactions. The Hate speech detector is built using the ULMFIT transfer learning model [115], whose classification performance on 3947 hate and non-hate instances collected from the Kaggle's detecting insult in social commentary training set, is evaluated based on the models' precision, recall, and F1 score. Preliminary model classification outcomes obtained show the detector achieving the highest performance scores in the non-hate speech classes. The most relevant features extracted during the hate speech classification phase with ULMFIT models are then leveraged and utilized in the next phase of the design process. The proposed design framework is optimized by adopting the feedback from the last iteration and reevaluating, adjusting, and updating the dashboard prototype with the relevant information retrieved earlier. These result in an interpretable module assimilated into the interface and capable of providing explanation to input text deemed hate related. The framework's explainability is supplied to users as visual highlights of the most impactful features flagged by the model. These highlighted features, computed and displayed as prediction probabilities during the modeling process, are also displayed on the systems' dashboards as confidence scores to the models' decisions about the text. Both prototypes (with and without explanation) are then tested and validated with respect to reviews and assessments from 100 Amazon M-Turk workers evaluating the system's overall perceived usefulness, ease of use, trustworthiness, intention, and usage experience on a Five-point Likert scale. The response collected from this phase, assessed and statistically analyzed following the Whitney U tests [116], shows that integrating explainable features and components into the overall design made it significantly more useful, trustworthy, understandable, and easier to use.

Similarly, [27] proposes an approach to speed up online discussion moderation by leveraging the robustness and classification powers of NN algorithms to generate an inappropriate comments detector capable of providing visual highlights of the critical cues responsible for the models' decision and prediction outcomes. The claim classification task is undertaken by training the detection algorithm on a proprietary collection of 20 million under-resourced Slovak new comments, focusing mainly on offensive content flagged as either racist, insults, profanity, or spam. The author experiments with multiple algorithms, including uni- and bi-directional Recurrent Network algorithms with RCNN and LSTM cells. The resulting best model for the task is selected through grid searching, and the models are evaluated based on the computed Accuracy and Average Precision (AP) scores. The bi-directional Recurrent net algorithm with RCNN cell performs best with accuracy and AP of 0.794 and 0.875, respectively, compared to 0.872 and 0.794 for its counterpart with LSTM cell. It is also observed that the former model is far less resource intensive, as it achieves these outputs with 8.5 times fewer parameters comparatively. A ROC curve is also implemented

here, to assess and visualize the model's ability to distinguish between the classes, and the results obtained corroborate the initial outcomes. The method's explainability involves highlighting inappropriate cues from the input data, and that entails implementing an approach capable of selecting rationales – words that most trigger the NN decision, even from a set of unannotated data. This involves analyzing the FastText word embeddings custom-built during the training phase to generate the following two outcomes: One that contributes to the creation of a set of rationales from the offensive contents, and another that provides the model's scores over the two classes. In addition to marking a comment as offensive, the proposed method also provides justifications for the models' decision, by highlighting the offensive cues in the comments with up to 90% precision, even on a sentence that has undergone significant word reduction of about 10–15%.

4.4 Explainability Approach for Surfacing Hate in Under-Resourced Languages

An explainable detection method—DeepHateExplainer, for surfacing hate speech from the Bengali language using ML, DL models, and BERT variants is presented in [106]. The proposed approach provides several steps: the creation and extension of the Bengali hate speech dataset1 resulting in a total of 8,087 examples categorized as either political, religious, geopolitical, or personal comments. The data is used to train a collection of interpretable ML baseline algorithms, including LR, SVM, KNN, NB, RF, and GBT, following a character n-grams and word Unigram feature extraction approach with TF-IDF BOW encodings. Additionally, three deep learning baseline models, namely CNN, Bi-LSTM, and Conv-LSTM, are implemented, utilizing the FastText embedding approach for vectorizing and encoding the instances into n-grams with reduced words sizes to enable models to easily capture syntactic and semantic cues. Both these ML and DL-based baseline models are trained using a five-fold CV approach. The next step involves assessing the explainability of more sophisticated transformer-based neural architectures, mainly the Monolingual Bangla BERT base, XLM-RoBERTa [117], the multilingual BERT(cased and uncased) models, using a majority voting-based representation to extract the best classification outcomes. The WeightWatchers approach is leveraged to help select the top three BERT Variant models to be ensembled. The main idea is to assess the opaqueness of these black box models and determine how human-level interpretability could be incorporated to make them more understandable and transparent. The performance of the transformer models is then assessed and compared against the interpretable ML and DL baseline models before and after including local and global explanations of the transformer models. The performance of each model is evaluated and reported in terms of their overall Precision, Recall, F1 score, and **MC (Mathias correlation)** coefficient with the ML baseline models trained on manually selected features following a Forest of Trees concept. The results show the ML and DL baseline models achieving poor hate statement prediction outcomes compared to initial results from the transformer-based models. The XLM-RoBERTa model achieves the highest individual performance for BERT variants, but an ensemble of the top three BERT variant models, mainly the XLM-RoBERTa, Bangla Bert-base, and the m-BERT uncased achieve outstanding classification rates across the board. The conv-LSTM and GBT were the best-performing DNN and ML baseline models, respectively. The classification outcomes reported also show that personal hate is the most accurately identified, while political hate is the most accurately classified category of hate speeches. The approach for encapsulation subjective interpretability relies on **Sensitivity Analysis (SA)** [118] and **Layer-Wise Relevance Propagation (LRP)** methods for providing local and global model explanation. Global explanations are provided through feature importance computed using the SHAP and ELI5 methods, while local explainability is supplied by identifying the most relevant features for individual predictions retrieved from the features' relevance score computed from the SA and relevance conservation LRP, visualizable on a heat map. The ERASER benchmark is equally utilized to measure the quality of the model explanation generated based on the models'

computed faithfulness i.e., the comprehensiveness and sufficiency scores. To measure explainability on the ERASER benchmark, only the top performing ML, DL, and BERT variants models, i.e., the GBT, Conv LSTM, and XML-Roberta respectively, are considered. Ultimately, the BERT variants achieve the highest faithfulness score, hence corroborating observations [104] about models' performance being a counter-indicator for their overall understandability.

A proposal for detecting social media hate speech in under-resourced languages from a different perspective is presented in [119]. In this study, classic ML models such as LR, Multinomial NB, **Random Forest Decision Tree (RFDT)**, and XGB are used to detect and classify multilabel hate speech and abusive language from Indonesian Tweets. The models' trustworthiness and interpretability are evaluated using LIME. The dataset used for this experiment is presented in [120] and comprises about 13,169 entries categorized as hate speech, abusive language, or both. The dataset contains relevant information about the demographics of individuals or groups to which the offense is directed, such as race, gender, religion, and disability status. Additionally, a subcategory emphasizes the degree of hate speech as either weak, moderate, or strong. The text data collected is cleaned and preprocessed, using word n-gram and the TF-IDF methods for feature extraction and vectorization. The output vectors are then fed to different models, and their overall performance is evaluated and compared based on their Accuracy and F1 score. The classification outcomes show all models achieving 0.79 F1 scores and 0.83 Accuracy scores except for RF models, which achieve 0.82 accuracy scores. The models' explainability is assessed by inspecting and comparing their LIME explanations over a small sample of the correctly predicted outcomes and then providing a visual representation of the model's modeling behaviors through text highlight. Overall, the XGB classifier provides the most logical explanation and emerges as the best model in both classification performance score and explainability. This research emphasizes the role of XAI models in choosing the most optimal models for a particular task from a slew of equally performing approaches.

Similarly, another investigation from [121] utilizes a collection of ML algorithms, namely XGB, RF, DT, Adaboost, and **Gradient Boosting Classifier (GBC)**, to train and classify language toxicity. The authors implement DL and NLP techniques with LIME for models' explainability. After the data preprocessing phase, which involves information cleaning, tokenization, stemming, and word implantation, the authors apply Word2vec, Glove, and FastText methods modeled with a CNN classifier on the Kaggle Jigsaw Toxic dataset to select the best word embeddings for the task. For this task, the FastText model outperforms all others and is used to generate the training vectors, which are subsequently applied to the five models. The models' performance is assessed based on their Precision, Recall, and F1 score, and by visualizing the models' learning and validation curves. A recursive feature elimination technique with a 10-fold CV is also implemented to compare model results. In the end, the models are ensembled using a soft voting classifier approach to generate the best classification outcome with a 0.997 accuracy score. The outcome of the ensemble model is explained by assessing the model explanation on the randomly selected toxic and non-toxic comments. The expandability investigation and analysis reveal that the ensemble method can produce 92% and 89% prediction probability for toxic and non-toxic comments, respectively. Overall, this study highlights the importance of proper text preprocessing techniques applied during modeling and demonstrates that the combined effects of CNN and quick text word embeddings on well-curated and processed data could result in SOTA classification outcomes, as well as an optimal hate speech detector's interpretability.

5 Discussion

This section presents an overview of XAI techniques extracted from this survey that have been applied to helping mitigate these issues. We present a summary of the datasets, the XAI

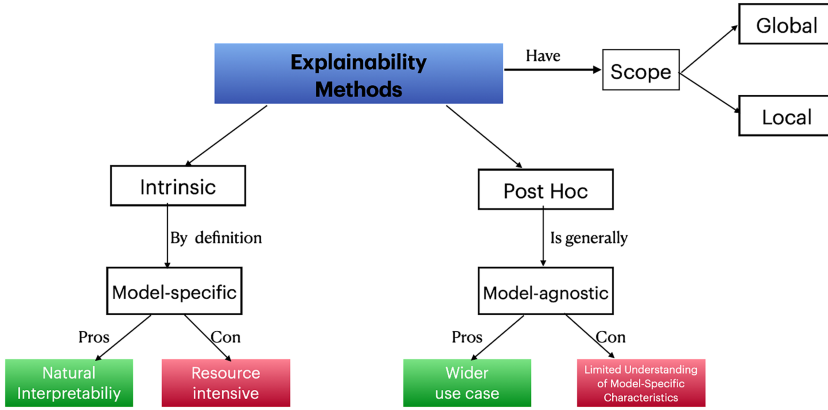


Fig. 6. Simplistic flow of XAI methods' taxonomy.

methods leveraged throughout the survey classified as either intrinsic or extrinsic, key observations, strengths, and limitations of our work as well as opportunities for future research.

5.1 Overview of XAI Techniques Applied to Hate Speech and Fake News Detection

Interpretability/explainability are essential factors in the creation and deployment of ML models and there exist two main approaches for incorporating them into models: Model-specific or Intrinsic methods, which have built-in interpretable elements while model-agnostic approaches embed explainability in the model architecture. While model-specific approaches provide natural explainability, they can be resource intensive, and it is this important limitation that has led to their decrease in popularity and increasing adoption of model-agnostic methods, which can provide explainability to a wide range of use cases comparatively. These methods are not specific to one ML model and can effectively separate models' classification outcomes from the explanation, as can be seen in Figure 6 which provides a simplistic overview of XAI models taxonomy as presented by [131]. Examples of intrinsic methods from our survey include LR, SVMs, KNN, Bayes, and tree-based models such as DT and RF, while the model-agnostic methods include use cases in outcome Visualization [62, 72, 106, 112, 119], Knowledge extraction techniques including Model distillation [29, 58], Influence methods such as SA, ranking and Layer wise relevance propagation [106], and feature importance computation [94]. For a detailed dive into the features, characteristics, and descriptions of these XAI methods, as well as some of the model explainability techniques highlighted, refer to research by Holzinger et al. [122]. In their work, the authors analyze 17 of the most common XAI methods while providing resources and links to GitHub repositories of the implementation of the XAI methods explored. At a foundational level, Ras et al. (2022) [123] research provides an initiation guide, focusing specifically on the explainability of deep Neural Nets while Gongane et al. (2023) [10] work highlights relevant examples, definitions, and categorization of XAI models scope and methods for fake news and hate speech. From a deep learning models perspective, we find that some of the most widely used models, like CNN and RNN, can be difficult to interpret. However, model-agnostic techniques have often been leveraged to provide explainability to a wide range of use cases, including deep NN models. Among these models, the RNN-based LSTM model has emerged as a popular choice with implementations in various ways, including as a comparative alternative to CNN models for stance classification [55], with attention mechanism [63, 137] to compute attention scores for instance explanations, creating explainability rank list and in combination with CNN to investigate attention coverage [72].

5.2 Overview of Datasets Extracted from the Survey

We have examined the effectiveness of XAI methods in detecting fake news and hate speech on social media platforms like X and Meta, as well as on news outlets with political content from popular sources such as Breitbart News, BBC News, CNN.com, The New Yorker, and others. Throughout this survey, we have underscored the importance of XAI in fostering ethics and trust in AI systems. However, the success of these methods is significantly influenced by the type and quality of datasets used in their development and implementation.

To support the ongoing development of ethical and trustworthy XAI applications in combating these social incivilities, we present a comprehensive summary of the datasets referenced in this survey. This includes details on each dataset's content type and language, origins, classification labels, and size, with direct links provided for accessibility. We believe this resource will be invaluable for the AI and NLP community. Refer to Table 2 of the supplementary material provided for the summary table.

5.3 Key Findings, Research Strengths, and Limitations

From the summary of Intrinsic and post hoc explainable models gathered from this survey and presented in Table 1, several assessments can be made regarding these explainability methods for fake news and hate speech detection. Notably,

- Post hoc methods are generally widely used for both fake news and hate speech detection.
- Intrinsic methods are more commonly applied to fake news detection than hate speech detection.
- Fewer studies use intrinsic methods for hate speech detection overall, which could indicate an open research direction for developing models that inherently understand and explain hate speech.

Moreover, some of the findings and conclusion presented in this work provides evidence dispelling some common misconceptions about model explainability. One such misconception is that model performance is the most important factor in determining its explainability. Following observation from experiments conducted by [113] and [115], we note performance does not always infer explainability. Another misconception is that a model's overall prediction accuracy is the most important factor in building trust in its outcomes. However, as demonstrated by [73], a model's explainability can also significantly contribute to human trust in its predictions. For a detailed overview of the tradeoff between model accuracy and interpretability, refer to research by Ali et al. (2023) [129]. We also observed that effective fake news and hate detection can be achieved without sacrificing transparency, and that fake news stories tend to contain emotionally charged content to attract readers. While researchers have provided cues on how to identify fake news on social media, there are still limitations and challenges in creating and deploying explainable fake and hate speech detectors. For instance, reliance on human rationales and annotations may perpetuate bias throughout the pipeline, while model explainability may slow down human assessment of a system, as reported in [29].

Overall, a comprehensive review of recent advancements in XAI techniques applied to the detection of misinformation and hate speech is presented in this manuscript. A summary of methods, datasets, evaluation metrics, and interpretability techniques used in the development of trustworthy and ethically fair detection models is also emphasized, thus contributing to the ongoing efforts to create more human-friendly models. We hope that the content of this review will serve as a useful resource for researchers and practitioners working in this field. Nonetheless, despite the relevance of the collection of datasets extracted, one important limitation is its non-diversity, as

Table 1. Intrinsic and Post Hoc Explainable Models Gathered from Papers on Hate Speech and Fake News Detection

Intrinsic Methods	Fake news	Shu et al. (2019) [63]	Yang et al. (2019) [62]	Lu et al. (2020) [73]	Belloir et al. (2022) [33]	Kurasinski and Mihailescu (2020) [72]	Ni et al. (2021) [81]			
	Hate Speech	Cindy Wang (2018) [94]	Svec et al. (2018) [27]							
Post hoc Methods	Fake news	Reis et al. (2019) [51]	Alharbi et al. (2021) [69]	Gadek and Guélorget (2020) [38]	Qiao et al. (2020) [45]	De Magistris et al. (2022) [55]	Vijjali et al. (2020) [57]	Mohseni et al. (2020) [58]	Denaux and Gomez-Perez (2020) [88]	Szczepański et al. (2021) [84]
	Hate speech	Wich et al. (2020) [111]	Wich et al. (2022) [112]	Mattheu et al. (2020) [104]	Mehta and Passi (2022) [105]	Ibrahim et al. (2022) [119]	Shakil and Alam (2022) [121]	Hardage and Najafirad (2020) [97]	Karim et al. (2020) [106]	Bunde et al. (2022) [113]

it comprises mostly English and other Western languages and does not account for other popular languages such as Urdu, Swahili, Creole, Amharic, and Zulu. We are confident that emphasizing and prioritizing the development and integration of diverse datasets, especially for languages that are currently under-resourced or underrepresented for most social incivilities’ detection, will be crucial in advancing the capabilities of XAI in detecting them across different linguistic and cultural contexts

5.4 Future Directions and Research Opportunities

AI is making decisions that impact our lives in countless ways. But what happens when these decisions are so complex that even the most skilled experts in the fields struggle to understand them? Comes the field of XAI, which is rapidly becoming an essential tool for building AI systems that are transparent, ethically fair, and thus responsible. XAI helps address critical issues such as bias, discrimination, and lack of transparency, by enabling better comprehension of the inner workings of AI systems, thus enabling identification of potential biases or errors that can lead to discriminatory outcomes. This is especially important for communities that have historically faced discrimination, as discussed in recent research [124–125]. Without XAI, these groups may be unfairly targeted or excluded from the benefits of AI technologies, especially in domains such as healthcare [126] where XAI would contribute to enhancing user trust in AI systems used in healthcare and strengthening doctor–patient relationships. From a media perspective, XAI plays a pivotal role in promoting digital literacy and media accountability by providing clear and understandable explanations to users about why certain content has been flagged, along with useful information related to their searches and responses. This enables users to critically evaluate the information they consume, fostering a more informed and discerning audience. These are some of the motivations behind our future work.

Our future endeavor aims to deepen our understanding of the specific characteristics of fake news and hate speech content by analyzing the emotional and sentiment-driven nature of misinformation to assess how it may increase, contribute to, or incite hate. This insight will be significant in tracking fake news and predicting peaks in hate speech caused by misinformation, with the goal of better anticipating the next wave of online or offline hate, thus helping effectively de-escalate and mitigate such incivilities. To that effect, Sentiment analysis methods like those

presented in [127] could be leveraged to facilitate our study. Furthermore, exploring the psychological mechanisms and emotional triggers that make individuals susceptible to both believing or spreading fake news and engaging in hate speech would be another crucial avenue to explore. Research should focus on cognitive biases, emotional contagion, and the role of social identity in these processes. For example, studies could examine how emotions such as fear and anger are manipulated to propagate hate-driven misinformation and how reinforcing critical thinking skills could counteract these effects. Thus, contribute to the creation of effective educational programs and digital literacy campaigns for mitigating their effects in society.

A key takeaway from this survey is the limited research on multimodal approaches. While our survey has primarily focused on text-based models, a promising future direction involves applying XAI to multimodal data, combining text, images, videos, and other data types. Notably, only one paper [108] effectively applies a multimodal approach leveraging users' tweets and network data, which following the author's outcomes, could be essential for tracking peaks in fake news that may incite hate, as these data sources provide real-time insights into how misinformation spreads and gains traction across social networks. The network data would be able to reveal the structure of information diffusion, identify key influencers who propagate misinformation, and uncover patterns in how different communities interact with and respond to fake news. The limited use of multimodal data in hate speech and fake news detection may stem from the complexity of the task or the computational challenges associated with handling diverse data sources. Future research should prioritize the development of more effective multimodal datasets and XAI methods to address this gap.

Moreover, a significant challenge in combating fake news and hate speech is the diversity in their expression across different languages, cultures, and domains. To address this, future research should focus on developing AI models that can generalize well across these variations. This involves not only training models on diverse datasets but also ensuring that they can adapt to new contexts without losing their effectiveness. Addressing this would require leveraging transfer learning, multilingual modeling, and cross-domain validation to create adaptable tools capable of recognizing and responding to varied expressions of misinformation and hate speech across varied domains.

Research into HITL explainability evaluation approaches presents another avenue for enhancing detection accuracy, systems reliability, and thus, user trust in AI systems. These approaches leverage the strengths of automated detection algorithms while integrating human judgment and expertise in various forms, as highlighted in [62, 63, 104, 105]. It has been shown in Section 4.2 that incorporating human annotation into the detection pipeline can enhance the user's credibility in the detectors' outcome. However, a significant drawback often reported is the lack of trained expertise. Strategies such as crowd-sourced annotation, Active Learning, and expert feedback mechanisms can be employed to mitigate these limitations. Additionally, developing user-friendly interfaces and guidelines for human annotators will facilitate the effective integration of HITL approaches into detection pipelines. Addressing this challenge could involve developing solutions, like the add-ons proposed by [84] and [88], or browser extensions or social media plugins capable of providing real-time feedback on the credibility of online content. While similar ideas have been implemented by X, this proposed avenue aims for a more generalized approach accessible across various online platforms. Such solutions would empower users to make informed decisions about the content they consume in real-time, thereby complementing automated detection efforts effectively.

Considering growing concerns surrounding the ethical use of AI, XAI methods can play a crucial role in ensuring regulatory compliance and ethical standards. By providing transparency into AI decision-making processes, organizations can demonstrate accountability and adherence to ethical

guidelines. This transparency is particularly important in sensitive domains such as healthcare, finance, and criminal justice, where the implications of AI-driven decisions are significant.

6 Conclusion

Through a comprehensive review of recent trends and methodologies, we have explored how XAI can be applied to enhance online and offline decorum in public discourse, community safety, and the development of effective policies to eradicate fake news and hate speech. Our research has expanded on the synergistic relationship between these two phenomena and demonstrated how one impacts the other. We analyzed recent trends in XAI methods for detecting fake news and hate speech, as well as recent laws and regulations that enable the application of XAI to address these issues.

Our findings indicate a higher propensity for applying intrinsic methods to fake news detection, while post hoc methods are widely used across both fake news and hate speech detection. We also addressed common misconceptions about the relationship between model performance and explainability.

This research highlights the critical role of XAI in understanding and mitigating the challenges of fake news and hate speech. As AI technologies become increasingly integrated into various aspects of society, ensuring that these systems are transparent and accountable is essential for promoting fairness and trust. Our survey underscores the importance of using XAI to provide clarity and insights into the decision-making processes of AI models, particularly in domains where bias and misinformation can have severe social implications.

Acknowledgments

“**Meŋ Siakne**” Means **Thank you** in **Yemba** [130] (Popular Language spoken in Western Cameroon).

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